

# FitBuddy

AI Fitness Plan Generator using Gemini Models

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## PROJECT SUBMISSION

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SmartBridge × Smart Internz

Key Points Summary Report

# Project Introduction

FitBuddy is a web-based application that uses AI to generate personalized workout plans and nutrition tips based on a user's fitness goals — such as weight loss, muscle gain, or general wellness. Built with FastAPI and integrated with Google's Gemini AI models, FitBuddy simplifies the process of creating structured and effective fitness routines.

The application takes in user information — age, weight, goal, and preferred workout intensity — and generates a customized 7-day fitness plan. Users can also submit feedback to update and refine their plan using AI, and a complementary nutrition or recovery tip is provided based on their selected goal.

With a clean user interface and database support via SQLite, FitBuddy ensures each user's data and generated plan are stored and accessible. It offers both API endpoints and an interactive HTML interface for ease of use.

## Key Scenarios

- **Plan Generation:** User submits name, age, weight, goal, and intensity → receives a personalized 7-day workout plan.
- **Feedback Loop:** User submits feedback (e.g. "more cardio", "more rest days") → plan is regenerated/refined via AI.
- **Nutrition Tips:** A concise, goal-specific nutrition/recovery tip is generated alongside the plan.
- **Admin Dashboard:** Coaches/admins view all users, and compare original vs. updated plans.

## Core Technologies

- **Backend:** FastAPI — routing, input validation, business logic.
- **AI Layer:** Google Gemini 1.5 Pro (workout generation & feedback updates) + Gemini Flash (fast nutrition tips).
- **Database:** SQLite + SQLAlchemy ORM.
- **Frontend:** HTML + Jinja2 templates, styled with CSS (Flexbox, responsive).

## Technical Architecture

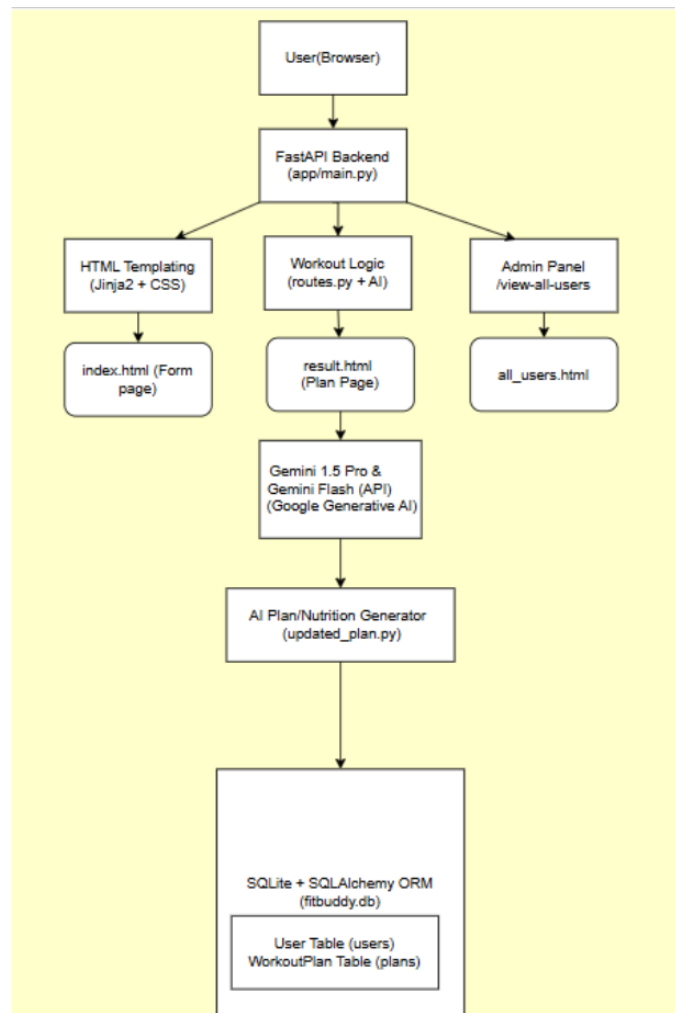


Figure: FitBuddy system architecture flow

### Why Gemini Models?

- Initially evaluated Hugging Face models (T5, BART, DistilBERT, Bloom) — required hosting/GPU setup and heavy fine-tuning.
- Selected Gemini for: easy API integration with FastAPI, native structured formatting, high accuracy with minimal prompting, and scalability.
- Gemini 1.5 Pro** → structured 7-day workout generation & plan updates.
- Gemini Flash** → fast, lightweight nutrition/recovery tips.

# Pre-requisites & Project Workflow

## Pre-requisites

- FastAPI framework knowledge
- Gemini API familiarity (Google Generative AI)
- HTML, CSS, and Jinja2 templating skills
- Python programming proficiency
- Git version control basics
- SQLAlchemy and SQLite fundamentals
- Environment setup (pip, virtualenv/conda) and Uvicorn ASGI server

## Project Milestones

| Milestone                         | Focus                                                            |
|-----------------------------------|------------------------------------------------------------------|
| 1. Model Selection & Architecture | Choose Gemini models, define architecture, set up environment    |
| 2. Core Functionalities           | Workout/nutrition generation, FastAPI routing & input processing |
| 3. routes.py Development          | Main application logic connecting frontend, AI, and DB           |
| 4. Frontend Development           | UI design, responsive CSS, Jinja2 dynamic templates              |
| 5. Deployment                     | Local deployment via Uvicorn, environment setup, testing         |

# Project Structure

create-frontend / app (python3.10) / templates (html) / static/images (assets)

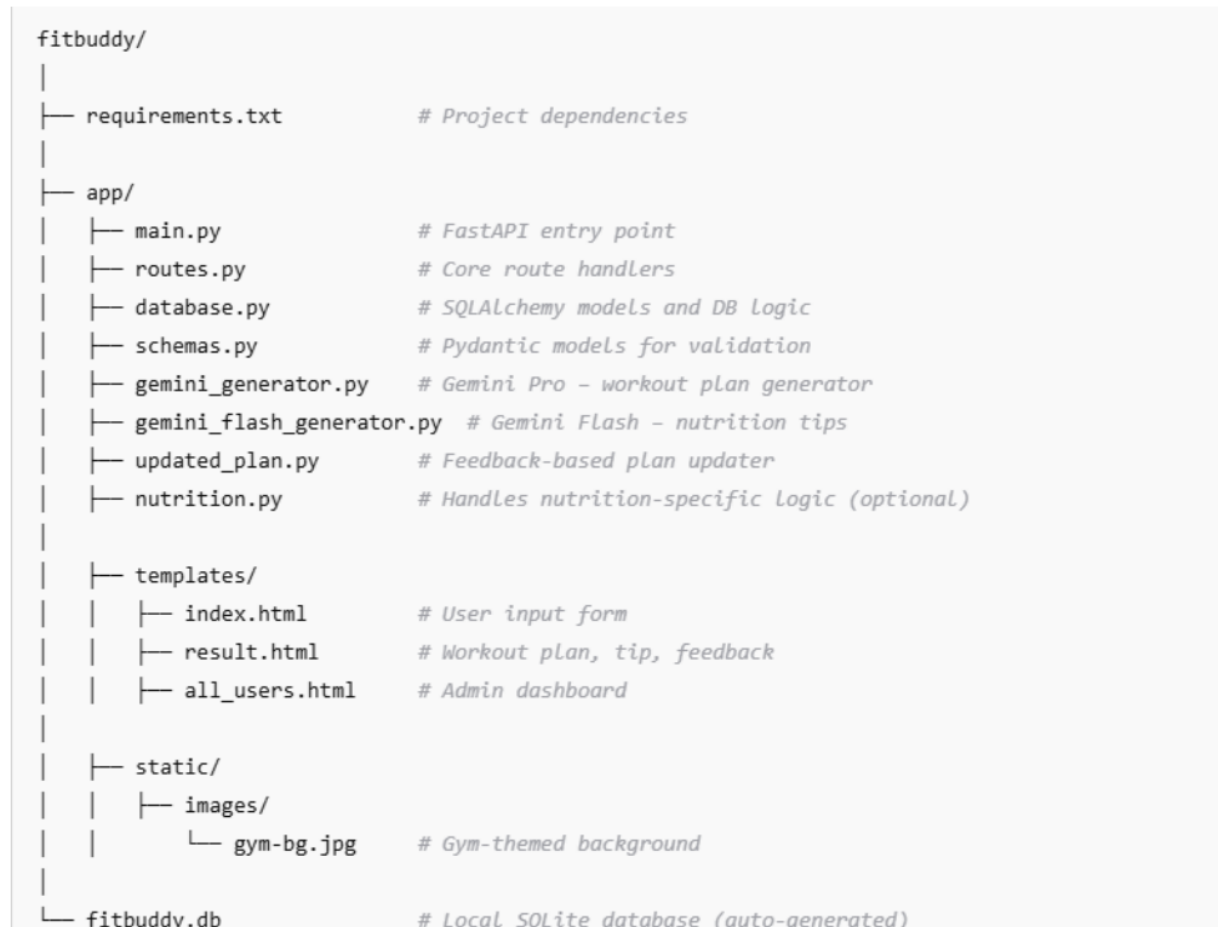


Figure: FitBuddy folder & file structure

## Structure Highlights

- **/app** — main.py (entry point), routes.py (route handlers), database.py (models/DB logic), schemas.py (Pydantic validation).
- **AI modules:** gemini\_generator.py, gemini\_flash\_generator.py, updated\_plan.py, nutrition.py.
- **/app/templates** — index.html, result.html, all\_users.html.
- **/app/static/images** — gym-themed background assets.
- **fitbuddy.db** — auto-generated local SQLite database.

## Core Functions

- **generate\_workout\_gemini()** (gemini\_generator.py) — creates a structured 7-day plan (warm-up, main workout, cooldown) via Gemini 1.5 Pro.
- **generate\_nutrition\_tip\_with\_flash()** (gemini\_flash\_generator.py) — generates a quick, practical nutrition/recovery tip via Gemini Flash.
- **update\_workout\_plan()** (updated\_plan.py) — revises the original plan based on user feedback via Gemini 1.5 Pro.
- **save\_user(), save\_plan(), update\_plan()** (database.py) — persist user info and plans (original & updated) to SQLite.
- **get\_original\_plan(), get\_user()** — retrieve stored user/plan data for feedback processing and admin view.

## Key API Routes

| Route                                      | Purpose                                                         |
|--------------------------------------------|-----------------------------------------------------------------|
| /                                          | Home page — displays index.html input form                      |
| /generate-workout or /generate-plan        | Processes input, generates & saves workout plan + nutrition tip |
| /submit-feedback or /update-plan/{user_id} | Updates plan based on user feedback via Gemini Pro              |
| /nutrition-tip                             | Generates a nutrition tip via Gemini Flash                      |
| /view-all-users                            | Admin dashboard — shows all users & their plans                 |

# Frontend Development

## Templates

- **index.html** — user input form (name, user\_id, age, weight, goal, intensity).
- **result.html** — displays the 7-day AI-generated workout plan, nutrition tip, and feedback form.
- **all\_users.html** — admin panel listing all users with original & updated plans.

## Design Highlights

- Gym-themed background image, Google Fonts (Roboto), bold fitness-oriented typography.
- Responsive layout using Flexbox and media queries; styled buttons with hover effects.
- Forms submit via POST to FastAPI endpoints (/generate-workout, /submit-feedback).
- Dynamic rendering via Jinja2 variables: {{ workout\_plan }}, {{ nutrition\_tip }}, {{ username }}, etc.
- Jinja2Templates configured in FastAPI to render TemplateResponse with context data.

## Home Page

home page.

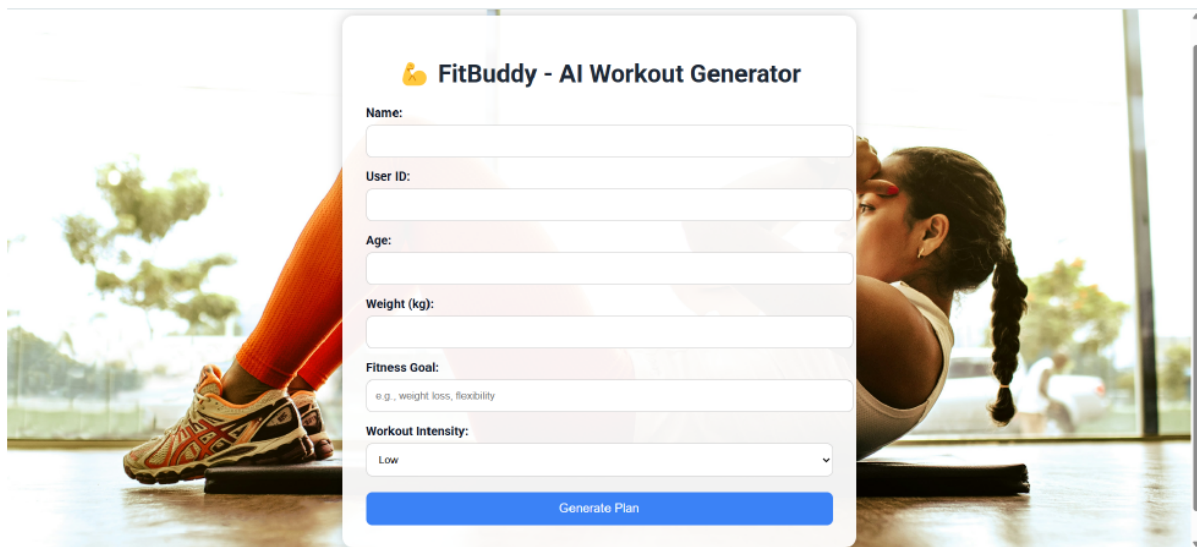
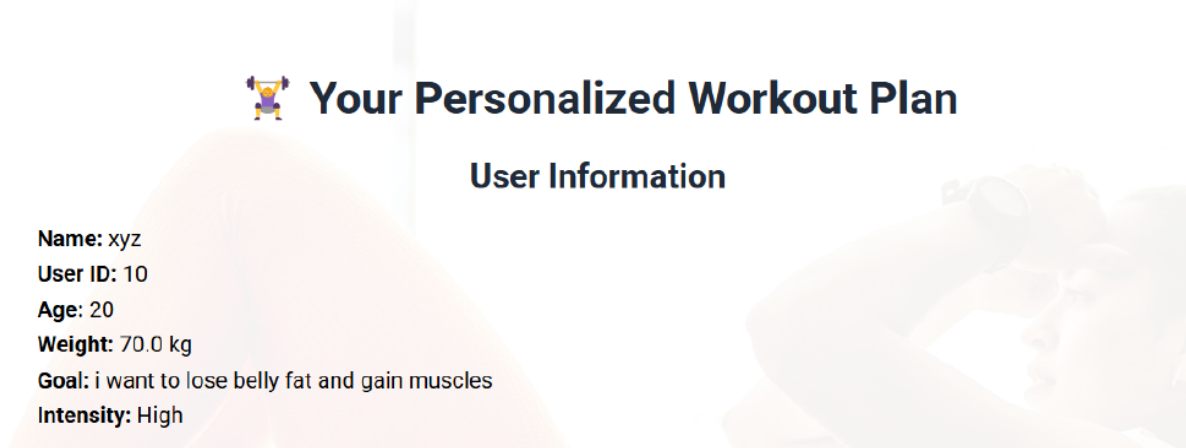
The image shows a web form titled "FitBuddy - AI Workout Generator" overlaid on a background image of a person in a gym. The form contains the following fields: "Name:" (text input), "User ID:" (text input), "Age:" (text input), "Weight (kg):" (text input), "Fitness Goal:" (text input with placeholder "e.g., weight loss, flexibility"), and "Workout Intensity:" (dropdown menu with "Low" selected). A blue "Generate Plan" button is at the bottom of the form.

Figure: FitBuddy homepage — user input form

# Application Walkthrough

## Personalized Workout Plan Page

After form submission, users see a summary of their input (Name, User ID, Age, Weight, Goal, Intensity), followed by the AI-generated 7-day plan and a nutrition tip.



The screenshot shows a web page titled "Your Personalized Workout Plan" with a dumbbell icon. Below the title is a section labeled "User Information". To the left of this section, a list of user details is displayed: Name: xyz, User ID: 10, Age: 20, Weight: 70.0 kg, Goal: i want to lose belly fat and gain muscles, and Intensity: High. The background of the page features a faint image of a person working out.

**Your Personalized Workout Plan**

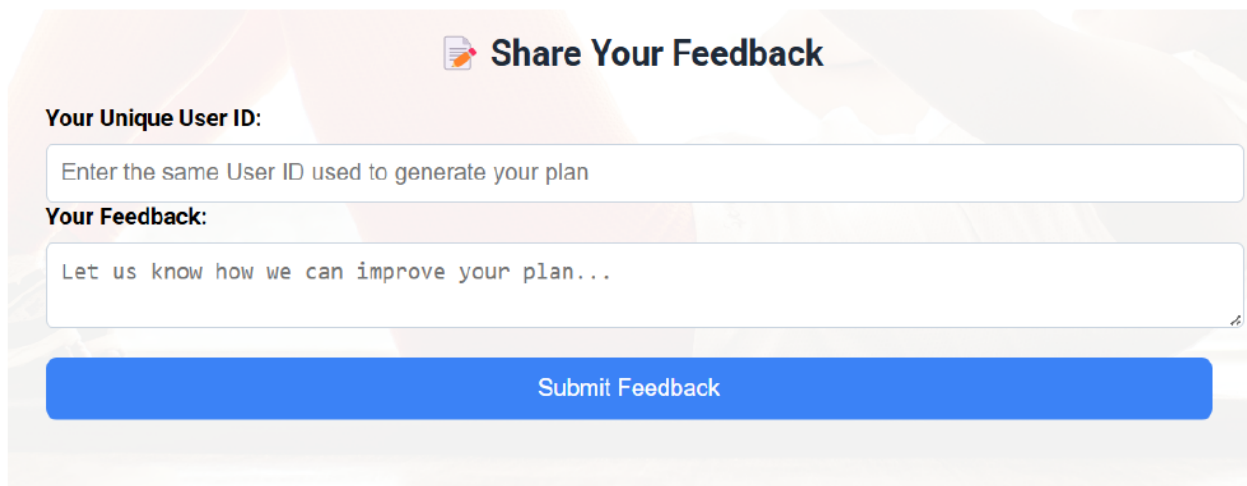
**User Information**

**Name:** xyz  
**User ID:** 10  
**Age:** 20  
**Weight:** 70.0 kg  
**Goal:** i want to lose belly fat and gain muscles  
**Intensity:** High

Figure: User information summary on the result page

## Feedback Page

Users can submit their User ID and feedback (e.g. "more cardio") to have their plan intelligently regenerated using Gemini 1.5 Pro.



The screenshot shows a web page titled "Share Your Feedback" with a document icon. It contains two input fields: "Your Unique User ID:" with a placeholder "Enter the same User ID used to generate your plan", and "Your Feedback:" with a placeholder "Let us know how we can improve your plan...". Below these fields is a blue "Submit Feedback" button. The background features a faint image of a person working out.

**Share Your Feedback**

**Your Unique User ID:**

Enter the same User ID used to generate your plan

**Your Feedback:**

Let us know how we can improve your plan...

**Submit Feedback**

Figure: Feedback submission form



## Admin Dashboard

The View All Users page gives admins/coaches an overview of the entire user database — personal details, original workout plan, and updated plan (if feedback was submitted) — enabling transparent progress tracking.

| FitBuddy - All Users & Workout Plans |        |     |             |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|--------|-----|-------------|-------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User ID                              | Name   | Age | Weight (kg) | Goal        | Intensity | Original Plan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Updated Plan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1                                    | shreya | 22  | 55.0        | muscle gain | high      | <p><b>## 7-Day Muscle Gain Workout Plan (Medium Intensity)</b></p> <p>This plan focuses on compound exercises to maximize muscle growth and utilizes a push-pull-legs split with rest days strategically placed for optimal recovery.</p> <p><b>**Day 1: Upper Body Push**</b></p> <p><b>**Warm-up (5-10 mins):**</b></p> <ul style="list-style-type: none"><li>* Arm circles (forward and backward, 1 min each)</li><li>* Shoulder rotations (1 min)</li><li>* Push-ups (as many reps as possible in 1 min)</li><li>* Light bench press (empty bar, 1 set of 10-12 reps)</li></ul> <p><b>**Main Workout:**</b></p> <ul style="list-style-type: none"><li>* <b>**Bench Press:**</b> 3 sets of 8-12 reps</li><li>* <b>**Incline Dumbbell Press:**</b> 3 sets of 8-12 reps</li><li>* <b>**Overhead Press:**</b> 3 sets of 8-12 reps</li><li>* <b>**Triceps Pushdowns:**</b> 3 sets of 10-15 reps</li><li>* <b>**Lateral Raises:**</b> 3 sets of 12-15 reps</li></ul> <p><b>**Cooldown:**</b> Static stretching focusing on chest, shoulders, and triceps (hold each stretch for 30 seconds).</p> <p><b>**Day 2: Lower Body (Legs &amp; Core)**</b></p> <p><b>**Warm-up (5-10 mins):**</b></p> <ul style="list-style-type: none"><li>* Bodyweight squats (10-12 reps)</li><li>* Lunges (5-6 reps per leg)</li><li>* Glute bridges (10-12 reps)</li><li>* Plank (30 seconds)</li></ul> <p><b>**Main Workout:**</b></p> | <p><b>## 7-Day Muscle Gain Workout Plan (Medium Intensity)</b></p> <p>This plan focuses on compound exercises to maximize muscle growth and utilizes a push-pull-legs split with rest days strategically placed for optimal recovery.</p> <p><b>**Day 1: Upper Body Push**</b></p> <p><b>**Warm-up (5-10 mins):**</b></p> <ul style="list-style-type: none"><li>* Arm circles (forward and backward, 1 min each)</li><li>* Shoulder rotations (1 min)</li><li>* Push-ups (as many reps as possible in 1 min)</li><li>* Light bench press (empty bar, 1 set of 10-12 reps)</li></ul> <p><b>**Main Workout:**</b></p> <ul style="list-style-type: none"><li>* <b>**Bench Press:**</b> 3 sets of 8-12 reps</li><li>* <b>**Incline Dumbbell Press:**</b> 3 sets of 8-12 reps</li><li>* <b>**Overhead Press:**</b> 3 sets of 8-12 reps</li><li>* <b>**Triceps Pushdowns:**</b> 3 sets of 10-15 reps</li><li>* <b>**Lateral Raises:**</b> 3 sets of 12-15 reps</li></ul> <p><b>**Cooldown:**</b> Static stretching focusing on chest, shoulders, and triceps (hold each stretch for 30 seconds), <b>**followed by 15-20 minutes of yoga focusing on poses that stretch the chest, shoulders, and triceps.**</b></p> <p><b>**Day 2: Lower Body (Legs &amp; Core)**</b></p> <p><b>**Warm-up (5-10 mins):**</b></p> <ul style="list-style-type: none"><li>* Bodyweight squats (10-12 reps)</li><li>* Lunges (5-6 reps per leg)</li><li>* Glute bridges (10-12 reps)</li></ul> |

Figure: Admin panel — all users & workout plans

## Deployment

- Set up virtual environment, install dependencies from requirements.txt.
- Store Gemini API key as environment variable (.env file with python-dotenv).
- Run server: **uvicorn app.main:app --reload**
- Access app at **http://127.0.0.1:8000**; API docs at **/docs**.
- Architecture is deployment-ready for future migration to cloud platforms.

## Conclusion

The FitBuddy – AI Fitness Plan Generator is a comprehensive, user-centric web application that showcases the powerful synergy between artificial intelligence and health-tech solutions. It uses Google's Gemini 1.5 Pro and Gemini Flash models to generate personalized 7-day workout plans and daily nutrition or recovery tips based on user inputs such as age, weight, fitness goal, and preferred workout intensity.

The application is built using the FastAPI framework for backend logic and API handling, SQLAlchemy with SQLite for persistent data storage, and Jinja2 templating for a clean and responsive frontend. What sets FitBuddy apart is its dynamic feedback feature — users can provide input on the effectiveness or difficulty of their current plan, and the system responds by intelligently updating the plan using generative AI.

Additionally, the application includes an administrative view where all registered users and their respective plans (original and updated) are displayed in a structured format — supporting transparency, monitoring, and further optimization for coaches or institutions managing multiple fitness profiles.

Overall, FitBuddy stands as a scalable and adaptable AI-powered solution that personalizes the fitness journey and evolves with the user's progress and preferences — demonstrating how cutting-edge technology can be integrated into everyday wellness routines.

## **Project Repository**

[https://github.com/mryogeshrajas-blip/FitBuddy\\_AI.git](https://github.com/mryogeshrajas-blip/FitBuddy_AI.git)

## **Demo Video**

<https://drive.google.com/file/d/1gqYg386Rg5VLzkGfdWVfz5mmCA-kKLrZ/view?usp=drivesdk>